

MET CS 555 – Foundation of Machine Learning

Module - 5

Lecture – 9: Anova and Regression, ANCOVA

Lecture – 10: One and Two- Sample Test of Proportion

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Anova and Regression

One-Way Analysis of Variance and Regression

We can use the linear regression to conduct the same tests in the one-way ANOVA setting

The one-way ANOVA is the **same as performing a regression** where the explanatory variable in the model is a variable or variables that indicate group membership.

In order to represent a one-way ANOVA model in the regression framework, construction of dummy variables is required.

For a particular categorical variable with **k categories, k-1 dummy variables are needed.**

$$\begin{aligned}\text{group}_2 &= \begin{cases} 1, & \text{if observation is in group 2} \\ 0, & \text{otherwise} \end{cases} \\ \text{group}_3 &= \begin{cases} 1, & \text{if observation is in group 3} \\ 0, & \text{otherwise} \end{cases} \\ &\vdots \\ \text{group}_k &= \begin{cases} 1, & \text{if observation is in group } k \\ 0, & \text{otherwise} \end{cases}\end{aligned}$$

One-Way Analysis of Variance and Regression

For a particular categorical variable with **k categories**, **k-1 dummy variables are needed**.

Dummy variables are binary variables of the form:

The category omitted (group 1 in this case) is referred to as the reference group.

A dummy variable for the reference group is not needed as those in this group can be identified as having values of 0 for all of the defined dummy variables.

$$\begin{aligned}\text{group}_2 &= \begin{cases} 1, & \text{if observation is in group 2} \\ 0, & \text{otherwise} \end{cases} \\ \text{group}_3 &= \begin{cases} 1, & \text{if observation is in group 3} \\ 0, & \text{otherwise} \end{cases} \\ &\vdots \\ \text{group}_k &= \begin{cases} 1, & \text{if observation is in group } k \\ 0, & \text{otherwise} \end{cases}\end{aligned}$$

One-Way Analysis of Variance and Regression

The following table summarizes how dummy variables would be defined for a grouping variable with k categories.

If you knew the values of each of the dummy variables for a particular observation, you would then know what group the observation was a member of.

Group	Dummy Variables			
	group ₂	group ₃	...	group _k
1	0	0		0
2	1	0		0
3	0	1		0
...				
k	0	0		1

Additional explanation and examples

Understanding dummy variables

Dummy variables are a way to include categorical data (like group membership) in a regression model.

How it works?

Imagine you have 3 groups (Group A, B, and C). To include them in a regression model, you need to create dummy variables to represent each group.

For a categorical variable with k groups, we only need $k-1$ dummy variables.

One group (say, Group A) is chosen as the *reference group*. The reference group is assigned a value of 0 for all dummy variables, while other groups are assigned values to distinguish them from the reference group.

Why $k-1$ and not k : If we had a dummy variable for each group, it would create redundancy (known as multicollinearity). The reference group provides a baseline for comparison.

Additional explanation and examples

Key insight:

Dummy variables: help us encode categorical groups in regression. For k groups, we use $k-1$ dummy variables, leaving one as the reference group.

Interpretation: each dummy variable coefficient in regression represents the difference between a group's mean and the reference group's mean.

Hypotheses and F-test: both ANOVA and regression test the same hypothesis—whether there's a significant difference between group means—using the F-test.

Practical application: you can use this approach to convert ANOVA problems into a regression framework, which might offer additional flexibility in analysis (e.g., adding other predictors).

One-Way Analysis of Variance and Regression

The following model can be used to compare means between groups in the regression framework:

$$y = \beta_0 + \sum_{i=2}^k \beta_{i-1} \text{group}_i + e$$

where

- ▷ y is the response or dependent variable. $\text{group}_2, \text{group}_3, \dots, \text{group}_k$ are the dummy variables that represent membership in groups 2, 3, ... and k , respectively.
- ▷ β_0 is the intercept (the sample mean in the reference group [group 1 in this case]).
- ▷ β_1 is the mean difference between group 2 and the reference group (group 1 in this case).
- ▷ β_{k-1} is the mean difference between group k and the reference group.
- ▷ e is the random error which we assume is normally distributed with a mean of 0 and a variance of σ^2

One-Way Analysis of Variance and Regression

We can use this model to test the hypotheses of interest in the one-way ANOVA setting.

We can use this model to test the typical ANOVA null hypothesis

$$H_0 : \mu_1 = \mu_2 = \dots = \mu_k$$

(All underlying population means are equal) against the alternative hypothesis

$$H_1 : \mu_i \neq \mu_j \text{ for some } i \text{ and } j$$

(At least two of the k underlying population means are different or not all of the underlying population means are the same/equal).

One-Way Analysis of Variance and Regression

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(All underlying population means are equal) against the alternative hypothesis

$$H_1 : \mu_i \neq \mu_j \text{ for some } i \text{ and } j$$

(At least two of the k underlying population means are different or not all of the underlying population means are the same/equal).

Mathematically, these hypotheses are equivalent to the global F-test hypotheses in regression where we test the null hypothesis that all slope coefficients are equal to 0 .

$H_0 : \beta_1 = \beta_2 = \dots = \beta_k = 0$ versus the alternative that at least one of the slope coefficients is different from zero

$$H_1 : \beta_i \neq 0 \text{ for at least one } i$$

Given that the **overall F-test for one-way ANOVA is equivalent to the global F-test for multiple linear regression (when dummy variables are used in the regression to represent group membership)**, the ANOVA table in each case is the same.

One-Way Analysis of Variance and Regression

$$H_0 : \mu_1 = \mu_2 = \dots = \mu_k$$

$$H_1 : \mu_i \neq \mu_j \text{ for some } i \text{ and } j$$

Mathematically equivalent to

$H_0 : \beta_1 = \beta_2 = \dots = \beta_k = 0$ versus the alternative that at least one of the slope coefficients is different from zero

$$H_1 : \beta_i \neq 0 \text{ for at least one } i$$

This also means that the **sum of squares between and the sum of squares within** from the **one-way ANOVA model** are equivalent to the regression **sum of squares and the residual sum of squares**, respectively.

Additional explanation and examples

Setting Up the regression model

$$y = \beta_0 + \beta_1 \times \text{Dummy B} + \beta_2 \times \text{Dummy C} + e$$

Interpretation:

β_0 : The average outcome (e.g., mean test score) for the reference group (group A).

β_1 : The difference in average outcome between group B and the reference group (group A).

β_2 : The difference in average outcome between group C and the reference group (group A).

e : The error term, representing the variation not explained by the model.

This model predicts the outcome for each group by comparing them to the reference group.

Real world example of One Way Anova Model

A researcher wants to compare the effectiveness of three diet plans (A, B, C) on weight loss. After 30 days, the weight loss (in kg) for participants in each group is recorded.

Data:

Group	Weight Loss (kg)
Diet A	2.4, 2.7, 2.9, 3.1, 3.0
Diet B	3.5, 3.7, 3.9, 4.1, 3.8
Diet C	4.8, 5.0, 5.2, 5.5, 5.4

Diet A, B, and C are categorical variables

In regression or ANOVA, categorical variables cannot be directly included because regression equations require numeric inputs.

To include diet in the model, we create dummy variables to represent the categories:

DietB: 1 if the participant is in Diet B, 0 otherwise.

DietC: 1 if the participant is in Diet C, 0 otherwise.

Diet A serves as the reference group, so it is not explicitly included as a dummy variable.

Interpretation:

β_0 : The mean weight loss for Diet A (the reference group).

β_1 : The difference in mean weight loss between Diet B and Diet A.

β_2 : The difference in mean weight loss between Diet C and Diet A.

Real world example of One Way ANOVA Model

Weight Loss = $\beta_0 + \beta_1 \times \text{DietB} + \beta_2 \times \text{DietC} + \epsilon$

DietB = 1 if the participant is in Diet B, and 0 otherwise.

DietC = 1 if the participant is in Diet C, and 0 otherwise.

Participant	Diet	Weight Loss (kg)	DietB	DietC
1	A	2.4	0	0
2	B	3.5	1	0
3	C	4.8	0	1
...	...	...	...	...

Fitted Model (Results):

Weight Loss = $2.82 + 0.98 \times \text{DietB} + 2.36 \times \text{DietC}$

Interpretation of Coefficients:

$\beta_0 = 2.82$: The average weight loss for Diet A (reference group).

$\beta_1 = 0.98$: Diet B leads to 0.980.980.98 kg more weight loss than Diet A.

$\beta_2 = 2.36$: Diet C leads to 2.362.362.36 kg more weight loss than Diet A.

Summary:

Diet is a categorical variable.

It is represented by dummy variables (DietB, DietC) in the regression model.

The reference category (Diet A) is not explicitly coded but serves as the baseline for comparisons.

Real world example of One Way ANOVA Model

Steps:

Hypotheses:

H_0 : the mean weight loss is the same for all diets ($\mu_A = \mu_B = \mu_C$).

H_a : at least one diet leads to a different mean weight loss.

Perform One-Way ANOVA:

Calculate group means:

Diet A: $\bar{x}_A = 2.82$

Diet B: $\bar{x}_B = 3.8$

Diet C: $\bar{x}_C = 5.18$

Perform ANOVA test (details omitted for brevity).

Results:

$F = 24.62$, $p < 0.001$: The means are significantly different.

Interpretation:

At least one diet leads to significantly greater weight loss. Post-hoc like Tucky analysis can determine which diets differ.

R functions for One-way ANOVA using lm()

Create dummy variables and One-way ANOVA using lm() function

```
# Create dummy variables
> data$g0 <- ifelse ( data$group==' Current heavy smoker', 1, 0)
> data$g1 <- ifelse ( data$group==' Current light smoker', 1, 0)
> data$g2 <- ifelse ( data$group==' Former smoker', 1, 0)
> data$g3 <- ifelse ( data$group==' Never smoker', 1, 0)

# One-way ANOVA using lm() function
> m2 <- lm( data$SBP~data$g0+data$g1+data$g2, data=data )
> summary ( m2)

> m3 <- lm( data$SBP~data$g1+data$g2+data$g3, data=data )
> summary ( m3)

> m4 <- lm( data$SBP~data$g0+data$g2+data$g3, data=data )
> summary ( m4)
```

The Golf Ball Example

In the golf ball example, the one-way ANOVA table was as follows:

	SS (Sum of Squares)	df (degrees of freedom)	MS (Mean Square)	F
Between	2583.3335	2	1291.6668	20.67
Within	750	12	62.5	
Total	3333.3335			

Let us we create dummy variables for 2 of the 3 brands, for example, for the Nike and Callaway brands, and ran a regression predicting distance from these two dummy variables.

The regression model in this case that is equivalent to the above one-way ANOVA model is given by

$$y = \beta_0 + \beta_{\text{Nike}} \text{group}_{\text{Nike}} + \beta_{\text{Callaway}} \text{group}_{\text{Callaway}} + e$$

where group_i is a dummy variable indicating whether or not the observation is of brand i .

The golf example: R commands

```
# creating dummy variables
> golf$g0 <- ifelse(brand=='Callaway', 1, 0)
> golf$g1 <- ifelse(brand=='Nike', 1, 0)
> golf$g2 <- ifelse(brand=='Titleist', 1, 0)

# Create the model
> m <- lm(dist ~ g1 + g2, data=golf)
> summary(m)

# pass the model to the anova function
> anova(m)
```

```
Call:
lm(formula = dist ~ g1 + g2, data = golf)

Residuals:
    Min       1Q   Median       3Q      Max
   -10      -5         0         5      10

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  285.000     3.536   80.61  < 2e-16 ***
g1          -25.000     5.000   -5.00  0.000309 ***
g2           5.000     5.000    1.00  0.337049
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 7.906 on 12 degrees of freedom
Multiple R-squared:  0.775,    Adjusted R-squared:  0.7375
F-statistic: 20.67 on 2 and 12 DF,  p-value: 0.0001297
```

The golf example: R commands

```
# creating dummy variables
> golf$g0 <- ifelse (brand=='Callaway', 1, 0)
> golf$g1 <- ifelse (brand=='Nike', 1, 0)
> golf$g2 <- ifelse (brand=='Titleist', 1, 0)
# Create the model
> m <- lm(dist~g1+g2, data=golf)
> summary(m)
# pass the model to the anova function
> anova(m)
```

Analysis of Variance Table

Response: dist

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
g1	1	2520.8	2520.8	40.333	3.66e-05 ***
g2	1	62.5	62.5	1.000	0.337
Residuals	12	750.0	62.5		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
 , |

Pairwise comparisons - t-test

If the F-test from ANOVA is significant, it suggests that at least one group mean is different. In other word, pairwise comparisons are post-hoc tests (Bonferroni, Tukey, etc) conducted after finding a significant F-statistic.

Pairwise t-tests are conducted to determine which specific group means differ.

Pairwise comparisons can also be performed in the regression framework. The most simple is the test of the underlying mean from group i to the reference group (group 1 as outlined above).

In the one-way ANOVA framework, the test of the null hypothesis $\mu_i = \mu_1$ would be accomplished through the use of the t statistic:

$$t = \frac{\bar{x}_i - \bar{x}_1}{\sqrt{s_p^2 \left(\frac{1}{n_i} + \frac{1}{n_1} \right)}}$$

which follows a t-distribution with $n-k$ degrees of freedom under H_0 (k here is the number of groups).

This test is the same as the t-test in the regression setting which tests the null hypothesis $\beta_i = 0$ after controlling for the other independent variables in the model.

Pairwise comparisons - t-test

$$t = \frac{\bar{x}_i - \bar{x}_1}{\sqrt{s_p^2 \left(\frac{1}{n_i} + \frac{1}{n_1} \right)}}$$

The t-statistic here is the same as the t-statistic for multiple regression:

$$t = \frac{\hat{\beta}_i}{\text{SE}_{\hat{\beta}_i}}$$

which follows a t-distribution with $n - k$ degrees of freedom under H_0

(k here represents the number of groups and not the number of variables in the model as we had defined it in the regression context).

Pairwise comparisons - t-test

It is also possible to perform other **pairwise comparisons** in the regression setting (for example, to test if there is a difference between groups i and j where neither i nor j represent the reference group), it **requires testing of null hypotheses of the form $\beta_i = \beta_j$ which are not standard** output from the regression model in most statistical packages and thus requires additional steps in order to calculate the test statistic for such comparisons.

- ▷ The same comparisons are relatively easy to make from the one-way ANOVA setting.
- ▷ If one were interested in using regression to make these comparisons, the **easiest way to do it is to run more than one regression analysis and change the reference group each time to a different group.**
- ▷ Obviously, conducting one-way ANOVAs gives this output standardly as such is often easier to implement in statistical software packages.

The golf example

For the golf ball distance by brand example, the below table summarizes the results of the **pairwise comparisons**. The group means were 285, 260 and 290 for Callaway, Nike, and Titleist, respectively.

Comparison	Mean Difference	<i>t</i>	df	Unadjusted p-value
Titleist versus Callaway	5	1.0	12	0.34
Titleist versus Nike	30	6.0	12	0.000062
Callaway versus Nike	25	5.0	12	0.00031

The summary of the output of the **regression model** below are show in the table:

$$\hat{y} = \beta_0 + \beta_{\text{Nike}} \text{group}_{\text{Nike}} + \beta_{\text{Callaway}} \text{group}_{\text{Callaway}}$$

	Estimate	SE	<i>t</i>	$Pr(> t)$
Intercept	290	3.536	82.02	< 0.0001
Nike versus Titleist	-30	5	-6	0.000062
Callaway versus Titleist	-5	5	-1	0.34

ANOVA vs Multiple Regression

Both ANOVA and multiple regression can be used to compare group means. They are closely related, but when using ANOVA model and when to use multiple regression?

1. Primary purpose of analysis

ANOVA: specifically designed to **compare the means** across multiple groups and **significant difference between** them. ANOVA focuses on testing for group differences,

Multiple Regression: generally more flexible and can include both continuous and categorical predictors. It is typically used for predicting a continuous outcome based on various predictors (not just group membership).

2. Simplicity for categorical data

ANOVA: since ANOVA was designed for comparing means across categorical groups, it provides a simple way to express this structure. When using one-way ANOVA, we're only concerned with a single categorical predictor, making the setup straightforward without needing multiple dummy variables.

Multiple regression: If we were to use multiple regression, we'd need to create dummy variables for each category, which can make the setup and interpretation more complex, especially when there are many categories.

ANOVA vs Multiple Regression

Both ANOVA and multiple regression can be used to compare group means, and they are closely related, but when using ANOVA model and when to use multiple regression?

3. Assumptions tailored for group comparisons

ANOVA assumptions: assumptions for ANOVA (normality within groups, homogeneity of variance, and independence of observations) are designed for analyzing group differences. The ANOVA model partitions **the variance** into “between-group” and “within-group” components, to address whether groups **differ significantly**.

Multiple regression assumptions: Assumptions for multiple regression are more general, as regression is designed to **predict** a continuous outcome.

4. Output and interpretation

ANOVA: anova provides a straightforward **F-test** that tells us if there is a significant difference between any group means. It's a single, global test to determine if at least one group is different from the others, which is often what researchers need in categorical comparisons.

Multiple regression: multiple regression provides t-tests for each coefficient (each dummy variable), which tells us if each individual group differs from the reference group. This can be useful if we need to compare each group individually to a baseline, but it can make interpretation more cumbersome if we're only interested in a global test of differences.

ANOVA vs Multiple Regression

Both ANOVA and multiple regression can be used to compare group means, and they are closely related, but when using ANOVA model and when to use multiple regression?

5. F-Test vs. Individual Coefficient Testing

ANOVA's F-test: the F-test in ANOVA gives a single p-value that tests whether there's any significant difference among all groups, simplifying the decision process.

Multiple regression's t-tests: each dummy variable in regression would have its own t-test, testing each group individually against the reference. While regression provides an overall F-test for the model, this is not always as directly interpretable in terms of group differences as ANOVA's F-test for group comparison.

6. Specialized for group comparisons

ANOVA: since it's designed specifically for comparing group means, it's often easier to communicate the results and interpret them in contexts where we only care about categorical group differences.

Multiple regression: more general-purpose and better suited for analyses where we're combining continuous and categorical variables or if we're interested in the relationship of each predictor with the outcome. If your analysis includes predictors beyond categorical groups, then multiple regression might be more appropriate.

Summary: ANOVA vs. Multiple Regression: When to Use

Use ANOVA:

When your primary goal is to test for **significant differences in group means** across categorical predictors.

It is best suited for scenarios with **one categorical predictor** or a straightforward comparison of group means.

Advantage: ANOVA provides a streamlined and focused analysis specifically tailored for testing group differences, without the need for dummy coding and interpreting coefficients.

Use Multiple Regression:

When your predictors are a mix of **categorical and continuous variables**, or when you are interested in understanding the individual contribution of multiple predictors.

It is more flexible than ANOVA, allowing for:

Interaction terms between predictors.

Adjustments for covariates.

Modeling more complex relationships.

ANOVA: Best for straightforward group comparisons.

Regression: Best for mixed predictors, complex models, or understanding specific predictor effects.

One-Way Analysis of Covariance **(ANCOVA)**

One-Way Analysis of Covariance (ANCOVA)

Sometimes we have **other variables or factors that have an effect on the dependent variables** that we are analyzing. But these variables are not actually independent variables.

You want to control the effect of other variables in your analysis.

For example: When you study the effect of different diet types on weight loss other factors like age or level of training can have an effect. You do not want to consider them as a separate independent variables, you just want to check if these variables have an effect on your analysis and control it.

We have multiple variables that have an effect on a single numeric outcome.

This is where we use **analysis of covariance**.

One-Way Analysis of Covariance (ANCOVA)

The other variables are variables that are called **“covariates”**.

These variables provide additional variations in the subject response value. We want to control these variables.

Covariates are also named “Intervening” or “Confounding Variables”.

Ancova is an extension of Anova and can be seen as a combination of Anova and Regression.

What ANCOVA does it simply takes out the effects of covariates out of Anova analysis.

One-Way Analysis of Covariance (ANCOVA)

- ▷ ANCOVA is a general linear model that is a **combination of ANOVA and regression** when you have some **categorical factors** and some **quantitative continuous variables**.
- ▷ The continuous variables (on which to perform regression) are called **“covariates”**
- ▷ Often these covariates are not necessarily of primary interest, but still their inclusion in the model will help explain more of the response, and hence reduce the error variance.
- ▷ ANCOVA evaluates whether population means of a response variable are equal across levels of a categorical explanatory variable, while statistically controlling for the effects of other continuous variables that are not of primary interest, known as covariates.
- ▷ Mathematically, ANCOVA decomposes the variance in the response variable into variance explained by the covariates, variance explained by the categorical explanatory variables, and residual variance.

One-Way Analysis of Covariance (ANCOVA)

The one-way ANCOVA model can be expressed as:

$$y = \beta_0 + \sum_{i=2}^k \beta_{i-1} \text{group}_i + \sum_{i=1}^j \beta_{k+i-1} x_i + e$$

where

- ▷ $\text{group}_2, \text{group}_3, \dots, \text{group}_k$ are the dummy variables that represent membership in groups 2, 3, ... and k , respectively.
- ▷ β_0 is the intercept (the sample mean in the reference group [group 1 in this case]).
- ▷ β_1 is the mean difference between group 2 and the reference group (group 1 in this case).
- ▷ β_{k-1} is the mean difference between group k and the reference group.
- ▷ β_k is the expected change in y for each one unit change in x_1 , across all groups and after adjusting for the other covariates $x_2, x_3, \dots, x_j$.
- ▷ β_{k+j-1} is the expected change in y for each one unit change in x_j , across all groups and after adjusting for the other covariates $x_1, x_2, \dots, x_{j-1}$

One-Way ANCOVA - Inferencing

In one-way ANCOVA framework, we are mainly interested in the overall model results as well as the results for the grouping factor (after adjusting for other variables in the model).

We can test the overall null hypothesis

$$H_0 : \beta_1 = \beta_2 = \dots = \beta_{k+j} = 0$$

(none of the variables in the model is predictive of the dependent variable y)

versus the alternative hypothesis

$$H_1 : \text{there is at least one } \beta_i \neq 0$$

(at least one of the variables is predictive of the dependent variable y)

One-Way ANCOVA - Inferencing

If the global null hypothesis is rejected, then we can test if the underlying population means are different across groups after controlling for the other variables in the model.

$H_0 : \beta_1 = \beta_2 = \dots = \beta_{k-1} = 0, \beta_k \neq 0, \dots, \beta_{k+j} \neq 0$ (all underlying population means are equal after controlling for $x_1, \dots, x_j$)

versus the alternative hypothesis

$H_1 : \text{At least two of the } k \text{ underlying population means are different}$
or not all of the underlying population means are the same/equal after controlling for $x_1, \dots, x_j$

One-Way ANCOVA - least-squares means or LS means

In the ANCOVA setting, if there is a difference between groups after adjusting for the other covariates in the model, we follow up with a comparison for the adjusted group means (instead of a comparison of the sample means).

The adjusted means are often called least-squares means or LS means.

- ▷ The **LS means** represent the **mean value for each group** that is adjusted for the other **covariates included in the model**.
- ▷ The **LS mean** in each group is calculated using the **least-squares regression equation** using the mean values of the covariates in the model.
- ▷ The general procedure and interpretation of the **one-way ANCOVA model** is similar to the procedure for the **one-way ANOVA**.
- ▷ The main difference is that in the ANCOVA setting, the inferences made are based on comparisons made after adjusting for or controlling for the covariates included in the model.

One-Way ANCOVA

```
# Don't use aov() function as it will not produce expected output
# Need to use Anova() function from package car to get Type
    III    sums of
square

> install.packages("car")
> library(car)
> Anova(lm(data$response ~ data$group + data$covariate), type=3)
```

car: Companion to Applied Regression

Functions and Datasets to Accompany J. Fox and S. Weisberg,
An R Companion to Applied Regression, Second Edition, Sage, 2011.

<https://cran.r-project.org/web/packages/car/index.html>

An example: One-Way ANCOVA

A random sample of current light smokers, current heavy smokers, former smokers, and those who have never smoked was taken to determine if mean systolic blood pressure (SBP) differs across smoking status categories.

Given that it is known that SBP increases with advancing age and given the fact that smoking preference are known to differ by age, we'd want to conduct an ANCOVA so that we can be sure that differences we see between smoking categories aren't due to differences in age.

It is preferable in this case to **use the ANCOVA methodology so that we can adjust for age.**

Another Real World Example for One-Way ANCOVA: Effectiveness of Training Programs

Does the training program (Program A vs. Program B) significantly affect the final test scores, after accounting for differences in baseline scores?

Employee	Baseline Score	Program	Final Score	ProgramB
1	60	A	80	0
2	55	A	75	0
3	70	A	85	0
4	50	B	70	1
5	65	B	78	1
6	60	B	72	1

What are variables:

1. **Baseline Score:** A continuous variable (covariate).
2. **Program:** A categorical variable with two levels (A, B). Coded as ProgramB1 for Program B, 0 for Program A.
3. **Final Score:** The dependent variable.

Model Setup:

In regression terms, ANCOVA fits the following model:

- $\text{Score} = \beta_0 + \beta_1 \cdot \text{Baseline} + \beta_2 \cdot \text{ProgramB} + \epsilon$

Another Real World Example for One-Way ANCOVA: Effectiveness of Training Programs

Our fitted model (results):

$$\text{Score} = 15 + 0.8 \cdot \text{Baseline} + 5 \cdot \text{ProgramB}$$

Intercept ($\beta_0=15$): The predicted score for Program A when the baseline is 0.

1. Baseline effect ($\beta_1=0.8$): For every 1-point increase in baseline, the final score increases by 0.8 points.

1. Program effect ($\beta_2=5$): Program B leads to a 5-point higher score than Program A, after accounting for baseline differences.

Another Real World Example for One-Way ANCOVA: Effectiveness of Training Programs

Steps:

1.Hypotheses:

1. Ho: No difference in final scores between program A and B after adjusting for baseline scores.
2. Ha: A significant difference exists between program A and B after adjustment.

2.Perform ANCOVA:

1. Covariate: Baseline Score.
2. Test for program effect on final scores, adjusting for the baseline.

Results:

Source	Sum of Squares	df	F	p-value
Baseline	74.30	1	21.50	0.01891
Program	32.06	1	9.28	0.05561
Residual	10.37	3		

Another Real World Example for One-Way ANCOVA: Effectiveness of Training Programs

Conclusion:

•Baseline Score ($p=0.01891$):

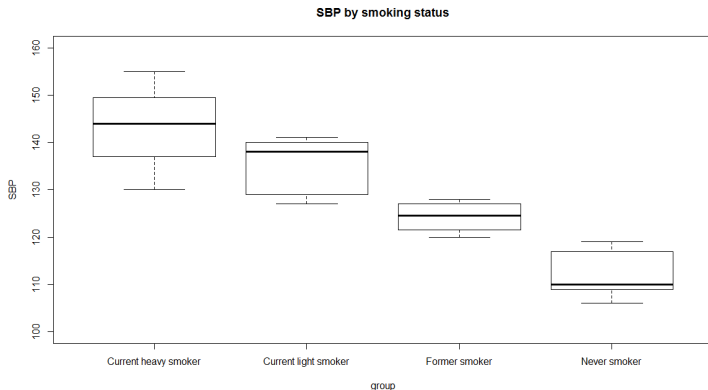
- Baseline Score has a significant effect on Final Score ($p<0.05$).
- For every 1-point increase in Baseline, the Final Score increases by 0.8 points.
- Baseline performance significantly predicts final scores ($p=0.0189$).
- Higher baseline scores are associated with higher final scores.

•Program ($p=0.05561$):

- The effect of Program is not significant ($p>0.05$).
- The adjusted difference between Program A and Program B is 5 points, but this difference is not statistically significant.
- After adjusting for baseline scores, there is no significant difference in final scores between program A and program B ($p=0.05561$).
- Although program B participants score 5 points higher on average, this difference is not statistically significant.

An example: One-Way ANCOVA

```
> data <- read.csv("smoking_SBP.csv")
> data
> aggregate(data$SBP, by=list(data$group), summary)
> boxplot(data$SBP~data$group, data=data, main="SBP by smoking status",
  xlab="group", ylab="SBP", ylim=c(100, 160))
```



An example: One-Way ANCOVA

First, ran a one-way ANOVA (without adjustment for age).

The **global F-test** showed that mean SBP differed by smoking **category** ($F=21.49$ on 3 and 15 degrees of freedom, $p < 0.001$).

```
> m<- aov( data$SBP~data$group, data=data )
> summary( m)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
data\$group	3	2786.2	928.7	21.49	1.1 e- 05 ***
Residuals	15	648.3	43.2		

Signif.	codes:	0	***	0.001	**	0.01	*	0.05	.	0.1
---------	--------	---	-----	-------	----	------	---	------	---	-----

```
> qf(.95, df1=3, df2=15)
[1] 3.287382
```


An example: One-Way ANCOVA

After adjusting for multiple comparisons using Tukeys methodology, all pairwise comparisons except for (a) between light and heavy current smokers and (b) between light and former smokers were significant at the $\alpha=0.05$ level.

```
> is.factor(data$grpnum)
> fnum = factor(data$grpnum)
> fnum
> m1<- aov(data$SBP~fnum, data=data)
> TukeyHSD(m1)
```

Tukey multiple comparisons of means
95% family-wise confidence level

```
Fit: aov(formula = data$SBP ~ fnum, data = data)
```

\$ fnum	diff	lwr	upr	p adj
1-0	-8.25000	-20.96090	4.4608964	0.2811214
2-0	-19.00000	-32.39846	-5.6015387	0.0047930
3-0	-31.41667	-43.64773	-19.1856009	0.0000119
2-1	-10.75000	-23.46090	1.9608964	0.1122314
3-1	-23.16667	-34.64042	-11.6929099	0.0001780
3-2	-12.41667	-24.64773	-0.1856009	0.0460500

An example: One-Way ANCOVA

However, after adjusting for age using an ANCOVA model, the differences seen in the one-way ANOVA setting were attenuated and the F-test for the **effect of smoking status was no longer significant** ($F=1.774$ on 3 and 14 degrees of freedom, $p=0.1982$).

The least square means (adjusted for age) were 129.87, 144.01, 144.01, and 155.11 for the heavy, light, former and never smokers, respectively.

As such, the differences that we saw in the one-way ANOVA model were due to age differences across the smoking groups as opposed to true differences in SBP attributable only to smoking status.

```
> is.factor(data$grpnum)
> fnum = factor(data$grpnum)
> fnum
> m1<- aov(data$SBP~fnum, data=data)
> TukeyHSD(m1)
```

One-Way ANCOVA

```
#Re-run ANOVA adjusting for Age
> library(car)
> Anova(lm(data$SBP~data$group+data$age), type=3)

# Least square means
# install.packages("emmeans")
library(emmeans)
my.model<-lm(SBP~group+age, data = data)
emm_options(contrasts=c("contr.treatment", "contr.poly"))
emmeans(my.model, specs = "group")

# no p value adjustment
emmeans(my.model, specs = "group" , contr = "pairwise", adjust="none")

# P value adjustment: tukey method
emmeans(my.model, specs = "group" , contr = "pairwise", adjust="tukey")

# P value adjustment: bonferroni method for 6 tests
emmeans(my.model, specs = "group" , contr = "pairwise", adjust="
  bonferroni")
```

One-Way ANCOVA

```
# Generate least square means (covariate adjusted means) and comparisons

> install.packages('lsmeans')
> library(lsmeans)
> options(contrasts=c("contr.treatment", "contr.poly"))
> lsmeans(lm(data$response ~ data$group + data$covariate), pairwise ~ data$
  group, adjust=[method])

# Note: method =  tukey ,  scheffe ,  sidak", "bonferroni",
  dunnett x",  mvt ",
"none")
```

lsmeans: Least-Squares Means

Obtain least-squares means for many linear, generalized linear, and mixed models. Compute contrasts or linear functions of least-squares means, and comparisons of slopes. Plots and compact letter displays.

<https://cran.r-project.org/web/packages/lsmeans/vignettes/using-lsmeans.pdf>

LSMeans adjusts the group means to account for covariates, such as age, by removing their influence. This ensures a fair comparison between groups by isolating the effect of the main predictor.

Two-Way Analysis of Variance

Two-Way Analysis of Variance

Sometimes we are interested in comparing the **mean response across groups** when there is **more than one factor** to be considered.

When there are **two factors** (each with two or more levels) that we are interested in, we use the **two-way ANOVA methodology** instead of the one-way ANOVA methodology.

The **goal of a two-way ANOVA is to look at the effects of each factor after controlling for the effects of the other factor.**

Two-Way Analysis of Variance

Within the two-way ANOVA framework, we seek to test the following:

- ▷ **Whether or not the first factor impacts the mean outcome** after controlling for the **second factor**.
Here, we test H_0 : All underlying population means are equal across levels of the first factor, after controlling for the second factor.
- ▷ **Whether or not the second factor impacts the mean outcome** after controlling for the first factor.
Here, we test H_0 : All underlying population means are equal across levels of the second factor, after controlling for the **first factor**.

One-Way ANOVA vs Two-Way ANOVA

One-Way ANOVA:

You have one categorical independent variable (factor) and want to test its effect on a dependent variable.

Hypotheses:

Null (H_0): All group means are equal ($\mu_1=\mu_2=\mu_3$).

Alternative (H_a): At least one group mean is different.

Formula:

$$Y = \mu + \alpha_i + \epsilon$$

Y: Dependent variable (e.g., test scores).

μ : Overall mean.

α_i : Effect of the i -th group (e.g., Method A, B, or C).

ϵ : Residual error.

Example: Do different diets (A, B, C) lead to different weight loss?

One-Way ANOVA vs Two-Way ANOVA

Two-Way ANOVA:

You have two categorical independent variables and want to test their individual and combined effects (interaction) on a dependent variable.

Hypotheses:

For Factor 1 (α_i): H_0 : No effect of the first factor (e.g., teaching method).

For Factor 2 (β_j): H_0 : No effect of the second factor (e.g., teaching style).

For Interaction ($(\alpha\beta)_{ij}$): H_0 No interaction between the factors.

Formula:

$$Y = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \epsilon$$

Y: Dependent variable (e.g., test scores).

μ : Overall mean.

α_i : Effect of the i-th level of the first factor (e.g., teaching method).

β_j : Effect of the j-th level of the second factor (e.g., teaching style).

$(\alpha\beta)_{ij}$: Interaction effect between the two factors.

ϵ : Residual error.

Example: Do different diets (A, B, C) and exercise levels (Low, High) influence weight loss, and does the effect of diet depend on exercise level?

Two-Way Analysis of Variance

The alternative hypothesis in each case is that the **underlying populations means are not equal across levels of the factor tested after controlling for the other**. At least two of the underlying group means are different across the factor tested, after controlling for the other factor.

Consider the following notation:

r = No. of levels of factor A.

c = No. of levels of factor B.

μ_{ij} = population mean for those in the *i*th level of factor A and the *j*th level of factor B.

$\mu_{i.}$ = population mean for those in the *i*th level of factor A across all *c* levels of factor B.

$\mu_{.j}$ = population mean for those in the *j*th level of factor B across all *r* levels of factor A.

		Factor B				Total
		1	2	...	c	
Factor A	1	μ_{11}	μ_{12}		μ_{1c}	$\mu_{1.}$
	2	μ_{21}	μ_{22}		μ_{2c}	$\mu_{2.}$
	...					
	r	μ_{r1}	μ_{r2}		μ_{rc}	$\mu_{r.}$
Total		$\mu_{.1}$	$\mu_{.2}$		$\mu_{.c}$	

To clarify further:

The i th and j th levels are essentially the values or categories within factor A and factor B.

Each combination of these levels represents a unique group or experimental condition in the study.

For example, in a simple experiment:

1. Factor A (e.g., Temperature):

1. Level 1 ($i=1$): "Low temperature"
2. Level 2 ($i=2$): "High temperature"

2. Factor B (e.g., Humidity):

1. Level 1 ($j=1$): "Low humidity"
2. Level 2 ($j=2$): "High humidity"

Temperature (Factor A)	Humidity (Factor B)	Outcome (μ_{ij})
Low ($i = 1$)	Low ($j = 1$)	μ_{11} : e.g., 50
Low ($i = 1$)	High ($j = 2$)	μ_{12} : e.g., 60
High ($i = 2$)	Low ($j = 1$)	μ_{21} : e.g., 70
High ($i = 2$)	High ($j = 2$)	μ_{22} : e.g., 80

.....

Here, the values in the table (μ_{ij}) are the outcomes (e.g., average plant growth, yield, or another response variable) for each combination of factor A (Temperature) and factor B (Humidity).

Two-Way Analysis of Variance

The global F-test is used to test the null hypothesis that there is no effect of either factor versus the alternative hypothesis that one of the factors has an effect **(there is an effect of either factor A or factor B or both)**.

The test for Factor A is testing the null hypothesis

$H_0 : \mu_{1.} = \mu_{.1} = \dots = \mu_{r.}$ against the alternative that

at least two of the underlying means are different $\mu_{i.} \neq \mu_{j.}$ for some i and j .

The test for Factor B is testing the null hypothesis

$H_0 : \mu_{.1} = \mu_{.2} = \dots = \mu_{.c}$ against the alternative that

at least two of the underlying means are different $\mu_{.i} \neq \mu_{.j}$ for some i and j .

Generally, the procedure for testing follows the procedure for the one-way ANOVA.

Interaction

Besides **the global test and the tests for each factor**, **a test of interaction must be performed** to understand if there is a non-additive effect of the either of the factors on the response.

One must first conduct an interaction test in the two-way ANOVA setting before the “**main effects**” type model can be conducted.

The concept of interactions is very important in statistics and in performing any type of multivariable analysis (including ANCOVA, multivariable regression, and two-way ANOVAs).

Interactions are when the effect of one factor is not constant over levels of another factor.

Interaction:

1. Interactions occur when the effect of one factor on the response variable depends on the level of another factor.
2. In other words, the relationship between a predictor and the outcome is not constant across levels of a second predictor.
3. For example, the effect of a treatment on recovery time might differ based on gender, meaning there's an interaction between treatment and gender.

1.Statistical Setting for Interaction Tests

1. **Two-way ANOVA** and similar models (e.g., ANCOVA, multivariable regression) allow for interaction terms to assess whether the combined effect of two factors is different from the sum of their individual effects.
2. For example, in a two-way ANOVA, an interaction test determines if the effect of factor A on the response varies depending on the level of factor B.

Why Are Interactions Important?

- Identifying complex relationships in the data.
- Accurately modeling real-world phenomena where factors rarely operate independently.
- Avoiding oversimplified interpretations of main effects when interactions exist.

Interaction

There is **an effect of the drug**, but **no effect of gender**.

- ▷ The presence and absence of an effect can be observed via the use of the **marginal means**.
- ▷ Those on drug B have a higher mean SBP (200) versus those on drug A (100).
- ▷ There does not appear to be an effect of gender (both males and females have an average SBP of 150).

		Drug		Total
		A	B	
Gender	F	100	200	150
	M	100	200	150
Total		100	200	

Interaction

There is **an effect of the drug** and **an effect of gender**.

There also appears to be an effect of gender (males have a higher SBP than females [200 versus 300, respectively]).

Here, the effects of gender are consistent across each drug and the effect of the drug are consistent across levels of gender. We conclude that though males tend to have higher SBPs than women, those on drug A have smaller SBPs than those on drug B. **In this case, there is not an interaction present.**

		Drug		Total
		A	B	
Gender	F	100	300	200
	M	200	400	300
Total		150	350	

Interaction

It appears as if there is **not an effect of drug or gender**, since all of the marginal means are equal.

On closer examination, it appears there is an effect of these factors, **but it varies based on the level of each factor**.

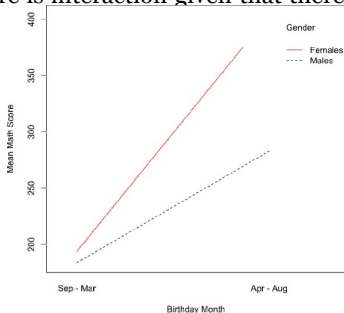
Because the effect of the drug depends on gender (and equivalently, the effect of gender depends on the drug used), **we say that these factors interact**.

		Drug		Total
		A	B	
Gender	F	100	300	200
	M	300	100	200
Total		200	200	

Interaction plots

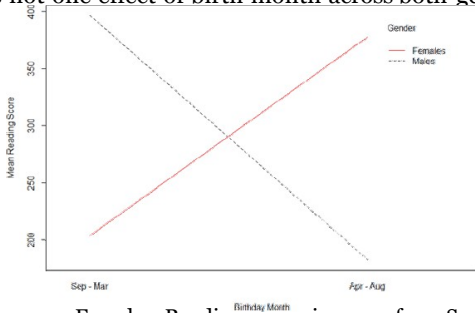
An educator is interested in the effect of birth month and gender on average reading and math scores for students in the fourth grade. A random sample is taken from each combination.

There is interaction given that there is not one effect of birth month across both genders.



Females: Math scores increase significantly from Sep-Mar to Apr-Aug.

Males: Math scores increase slightly but remain much lower than females across both birth month groups. The effect of birth month on math scores is much stronger for females than for males, suggesting an interaction.



Females: Reading scores increase from Sep-Mar to Apr-Aug.

Males: Reading scores decrease from Sep-Mar to Apr-Aug. There is a clear interaction where the effect of birth month on reading scores is opposite for males and females, as seen by the crossing lines (qualitative).

Interaction

If there is an interaction between two factors, then the conclusion that we'd get from the **two-way ANOVA are incorrect** at worst and misleading at best.

Before you perform a two-way ANOVA, **you must first check if an interaction is present.**

You do this by including an interaction term in the two-way ANOVA model.

If the **p-value of the interaction term is significant, the two-way ANOVA procedure is not appropriate.** In this case, one **must stratify the analysis by one of the factors and perform separate one-way ANOVAs** for the second factor for each level of the first.

If the interaction p-value is not significant, then one can proceed with the typical two-way ANOVA procedure and assume that the effect of each factor is consistent over the other.

Interaction

Given the importance of detecting interactions and the difference in interpretation of results in the presence of an interaction, alpha levels of 0.10 are often used to check if there is an interaction (as opposed to the more common 0.05 level of significance used in most other settings).

Two-Way ANOVA Example: Marketing Strategy and Region

We're examining the effects of Marketing Strategy (Online vs. In-store) and Region (North, South, East) on sales.

What are variables:

Observation	Region	Strategy	Sales	InStore	South	East	InStore x South	InStore x East
1	North	Online	15	0	0	0	0	0
2	South	Online	20	0	1	0	0	0
3	East	In-store	14	1	0	1	0	1
...	...	...	...	...	...	...	...	...

1. Marketing Strategy:

- Categorical with two levels: Online (reference group) and In-store.
- Dummy variable:
 - InStore=1 for In-store strategy, 0 otherwise

2. Region:

- Categorical with three levels: North (reference), South, and East.
- Dummy variables:
- South=1, 0 otherwise.
- East=1, 0 otherwise.

Regression Model (Two-Way ANOVA):

$$\text{Sales} = \beta_0 + \beta_1 \times \text{InStore} + \beta_2 \times \text{South} + \beta_3 \times \text{East} + \beta_4 \times (\text{InStore} \cdot \text{South}) + \beta_5 \times (\text{InStore} \cdot \text{East}) + \epsilon$$

Two-Way ANOVA Example: Marketing Strategy and Region

Fitted Model (Results):

$\text{Sales} = 12 + 3.5 \cdot \text{InStore} + 5 \cdot \text{South} + 6 \cdot \text{East} - 1.2 \cdot (\text{InStore} \cdot \text{South}) + 0.5 \cdot (\text{InStore} \cdot \text{East})$

Interpretation:

Intercept ($\beta_0=12$):

The average sales for the reference group (North region, Online strategy).

Main Effects:

$\beta_1=3.5$: Switching from Online to In-store increases sales by \$3.5k.

$\beta_2=5$: Sales in the South region are \$5k higher than in the North.

$\beta_3=6$: Sales in the East region are \$6k higher than in the North.

Interaction Effects:

$\beta_4=-1.2$: The impact of In-store strategy is \$1.2k lower in the South compared to the North.

$\beta_5=0.5$: The impact of In-store strategy is slightly higher (\$0.5k) in the East compared to the North.

Two-Way ANOVA Example: Marketing Strategy and Region

Steps:

Hypotheses:

H0: No effect of marketing strategy or region on sales.

Ha: There are effects of marketing strategy, region, or interaction.

Perform Two-Way ANOVA:

Main effects:

Marketing strategy (Online vs. In-store).

Region (North, South, East).

Interaction effect: Does the impact of the marketing strategy depend on the region?

Output:

Marketing strategy effect: $F=35.2$, $p<0.001$: Online performs better.

Region effect: $F=8.7$, $p=0.003$: Regions differ in sales.

Interaction: $F=2.1$, $p=0.12$: No significant interaction.

Interpretation:

Online marketing generates higher sales across all regions. Sales also differ by region, but the marketing strategy effect is consistent.

Two-Way Analysis of Variance

```
# Use Anova() function
> Anova([ model], type=3)

# First test interaction model
> model = lm(data$response ~ data$group1+ data$group2+ data$group1* data
  $group2)

# Visualize relationship using interaction.plot() function
> interaction.plot(data$group1, data$group2, data$response, col=1:2)

# If p-value for the interaction is not significant, then run regular
  two-way ANOVA
> model = lm(data$response ~ data$group1+ data$group2)
```


An example: Two-Way Analysis of Variance

An exercise physiologist wants to examine whether stretching and wearing ankle weights affect the value of exercise on treadmills. To carry out her study, she recruits subjects who have roughly the same level of physical fitness, and divides them randomly into four groups: with or without ankle weights with or without a stretching period before the exercise.

Using the amount of calories burned as the response, carry out a two-way ANOVA to determine whether stretching and wearing ankle weights have significant effects on exercise.

These are the variables in the data set:

Name	Type	Description
PreStretch	char	stretch group (Stretch, No stretch)
AnkleWeights	char	weights group (Weights, No weights)
Energy	num	calories burned
Speed	num	average speed (in meters per minute)
Oxygen	num	oxygen consumed (in liters)

An example: R commands

```
# Exercise example
exercise <- read.csv("exercise.csv")
exercise

attach(exercise)

#Test interactions
model <- lm(Energy ~ PreStretch + AnkleWeights + PreStretch * AnkleWeights, data
            =exercise)
summary(model)
Anova(model, type=3)

model1 <- lm(Speed ~ PreStretch + AnkleWeights + PreStretch * AnkleWeights, data
            =exercise)
summary(model1)
Anova(model1, type=3)

model2 <- lm(Oxygen ~ PreStretch + AnkleWeights + PreStretch * AnkleWeights,
            data=exercise)
summary(model2)
Anova(model2, type=3)
```

An example: R commands

```
# Generate interaction plots
interaction.plot(PreStretch, AnkleWeights, Energy, col=1:2)
interaction.plot(PreStretch, AnkleWeights, Speed, col=1:2)
interaction.plot(PreStretch, AnkleWeights, Oxygen, col=1:2)

# If interaction is significant, need to stratify (by more of the two
  factors)
stretch <- exercise[ which(PreStretch=='Stretch'),]
nostretch <- exercise[ which(PreStretch=='No stretch'),]

summary(aov(Energy~AnkleWeights, data=stretch))
summary(aov(Energy~AnkleWeights, data=nostretch))
summary(aov(Speed~AnkleWeights, data=stretch))
summary(aov(Speed~AnkleWeights, data=nostretch))
summary(aov(Oxygen~AnkleWeights, data=stretch))
summary(aov(Oxygen~AnkleWeights, data=nostretch))
```

MET CS 555 – Foundation of Machine Learning

Module - 5

Lecture – 10: One and Two- Sample Test of Proportion

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1. One-Sample Tests for Proportions
2. Two-Sample Tests for Proportions

One-Sample Tests for Proportions

One-Sample Tests for Proportions

- ▷ We are interested in the proportion of the population, p , that has particular outcome.
- ▷ The population parameter, p , is unknown and we aim to estimate it.
- ▷ As in previous sections, we seek to **make conclusions about the population parameter (p in this case) using information from a sample.**

The sample proportion, denoted $\hat{p}$, is an estimate of the population parameter, p .

- ▷ We look at the sampling distribution of the sample proportion to see how well it estimates the population proportion.

An Example: One-Sample Tests for Proportions

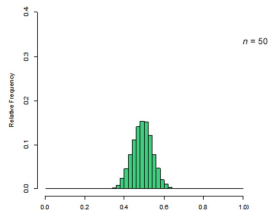
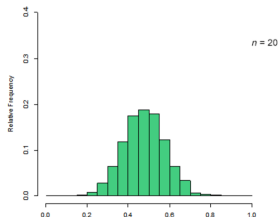
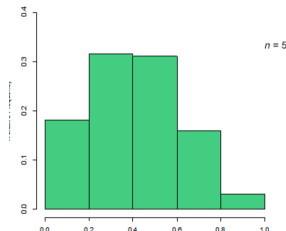
We are trying to estimate the proportion of all dice rolls that result in a value >3 . We'd like to estimate how often the die will roll a 4, 5, or 6.

For simplicity, we will call rolling 4, 5, or 6 a success.

We know that the probability of “success” using a fair die is $3/6$ or $1/2$ or 50%. Thus the underlying population parameter, p , is 0.50.

Let's assume we roll a die 5 times. In the 5 times, we get a 2, 2, 5, 6, and 3.

In this sample, we get $2/5=40\%$ as our sample proportion of “successes”.



One-Sample Tests for Proportions

We look at the sampling distribution of $\hat{p}$ as sample size increases:

1. As the sample size increases, **the sampling distribution of $\hat{p}$ becomes approximately normal.**
2. **The mean of the sampling distribution is p .**
3. **The standard deviation of the sampling distribution decreases as the sample size increases.** In fact, the variability (as measured by the standard deviation) of $\hat{p}$ is given by

$$\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}}$$

These properties mimic the properties of the sampling distribution of the sample mean as discussed in relation to the central limit theorem.

The Central Limit Theorem: When the number of samples taken from a population is sufficiently large, the sampling distribution of the sample mean, $\bar{x}$, will be approximately normally distributed with an expected value of μ and a standard deviation $\frac{\sigma}{\sqrt{n}}$.

A Few Caveats

Given the nature of the data being dichotomous in this setting, there are a few caveats.

- ▷ In general the normal distribution only holds **when the sample size is sufficiently large.**
- ▷ The given formula for the standard deviation of $\hat{p}$ is only accurate **when the population is much larger than the sample (at least 10 times larger)**
- ▷ Further, the normal approximation of the sampling distribution of $\hat{p}$ **works poorly when the underlying population parameter p is close to 0 or close to 1.**
- ▷ **When p is close to 0 or 1, much larger samples are required, why?**

If p is very close to 0 or 1, the distribution of $\hat{p}$ becomes skewed, and the normal approximation is no longer valid without a large sample size.

Significance Tests for a Proportion

Tests of hypotheses about a population proportion p are based on the sampling distribution of the sample proportion $\hat{p}$ and use data from a sample to evaluate evidence against the null hypothesis.

We are interested testing the null hypothesis $H_o : p = p_o$. To do this, we use the following z-statistic

$$z = \frac{\hat{p} - p_o}{\sqrt{\frac{p_o(1-p_o)}{n}}}$$

where

- ▷ $\hat{p}$ is the sample proportion,
- ▷ p_o is the value of the population proportion under the null hypothesis
- ▷ n is the number of observations in the sample.

This test is valid if the sample size n is sufficiently large. In general, we use the condition that inference is valid if np_o and $n(1 - p_o)$ are both greater than 10.

Significance Tests for a Proportion

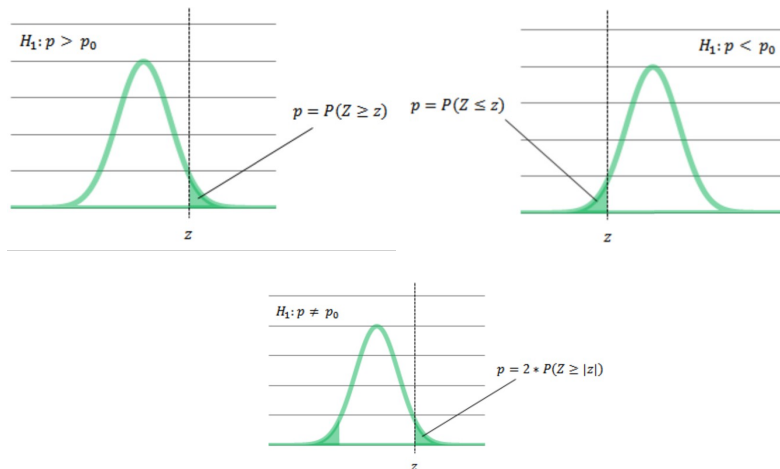
$$z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

- ▷ The test statistic for tests relating to the population proportion **measure how far $\hat{p}$ is from the value of p (under the null hypothesis)** in standard deviation units.
- ▷ **This result allows us to quantify** how far the point estimate is from the expected value of the population parameter under the null hypothesis and to make inference about the population proportion.

Significance Tests for a Proportion

This is generally done by calculating the corresponding p-value.

The p-value for the z statistic described above is calculated using the standard normal distribution.



An Example: Significance Tests for a Proportion

We are interested in estimating the proportion of children in the county that are vaccinated for measles. We suspect that it may be as low as 80%. A random sample is taken of 100 children from the county. Of those sampled, only 70 were vaccinated. Formally test if the proportion of vaccinated children is different than 80%.

1. Set up the hypotheses and select the alpha level

$H_0 : p = 0.80$ (the underlying population proportion is 80%)

$H_1 : p \neq 0.80$ (the underlying population proportion is different than 80%)

$\alpha = 0.05$

2. Select the appropriate test statistic

$$z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

An Example: Significance Tests for a Proportion

3. State the decision rule

Determine the appropriate critical value associated with a right hand tail probability of $\alpha = 0.05/2 = 0.025$. The critical value is 1.960. Decision Rule: Reject H_0 if $|z| > 1.96$ Otherwise, do not reject H_0

4. Compute the test statistic and the associated p-value

$$z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}} = \frac{\frac{70}{100} - 0.80}{\sqrt{\frac{\frac{80}{100}(1-\frac{80}{100})}{100}}} = \frac{-0.10}{\sqrt{0.0016}} = -2.50$$

5. Conclusion

Reject H_0 since $|-2.50| \geq 1.960$.

We have significant evidence at the $\alpha = 0.05$ level that $p \neq 0.80$ ($p = 2 * 0.0062 = 0.0124$). **We reject the null hypothesis that the proportion of children in the county that is vaccinated is 80%.** In our sample, just 70% of children were vaccinated.

Confidence Intervals for a Proportion

Confidence interval with a confidence level of C for the population proportion, p can be calculated by:

$$\hat{p} \pm z \cdot \sigma_{\hat{p}}$$

where

- ▷ $\hat{p}$ is the sample proportion
- ▷ z is the appropriate critical value corresponding to the confidence level
- ▷ $\sigma_{\hat{p}}$ is the standard deviation of $\hat{p}$.

However, $\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}}$ and p (the population proportion) is not known (and if it was, we wouldn't need to calculate a confidence interval!). We must instead estimate the quantity.

$$SE_{\hat{p}} = \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

The formula for the confidence interval for the population proportion, p , is

$$\hat{p} \pm z \cdot SE_{\hat{p}} = \hat{p} \pm z \cdot \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

An example: Confidence Intervals for a Proportion

Continue with the measles vaccination example.

The 95% confidence interval is calculated as follows:

$$\begin{aligned}\hat{p} \pm z \cdot \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} &= \frac{70}{100} \pm 1.960 \cdot \sqrt{\frac{\frac{70}{100} \left(1 - \frac{70}{100}\right)}{100}} \\ &= 0.70 \pm 1.960 \cdot \sqrt{0.0021} \\ &= 0.70 \pm 0.09 \\ &\approx (0.61, 0.79)\end{aligned}$$

We are 95% confident that the true proportion of vaccinated children in the county is between 61% and 79%.

One Sample Tests for a Proportion: R commands

```
# Use the prop.test() function
> prop.test([s], [n], p=[p0], alternative=[alternative], conf.level=[
  confidence
level], correct=FALSE)
# [s]: number of successes
# [n]: sample size
# [p0]: population proportion p
# [alternative]: two.sided (default), greater, less
# [confidence level]: default 0.95
# Correct = FALSE specifies not to use a continuity correction;
# Default is to apply correction
```

Continuity correction corrects for the fact that the sampling distribution of a proportion is not a continuous distribution. The correction should only be used when a single sample proportion is compared with the population proportion.

"Continuity correction is used only if it does not exceed the difference between sample and null proportions in absolute value." (Insofar as s/n does not exceed p_0)

It is a very slight alteration of the formula for the test statistics and the confidence interval that makes the test a little more conservative (and thus the confidence interval a little wider). **It is especially helpful when the sample size is small.**

An Example: One Sample Tests for a Proportion

```
> prop.test(70, 100, p=0.8, conf.level=0.95, correct=FALSE)
1-sample proportions test without continuity correction
```

```
data: 70 out of 100, null probability 0.8
X-squared = 6.25, df = 1, p-value = 0.01242
alternative hypothesis: true p is not equal to 0.8
95 percent confidence interval:
 0.6041515 0.7810511
sample estimates:
      p 
0.7
```

```
> prop.test(70, 100, p=0.8, conf.level=0.95, correct=TRUE)
1-sample proportions test with continuity correction
```

```
data: 70 out of 100, null probability 0.8
X-squared = 5.6406, df = 1, p-value = 0.01755
alternative hypothesis: true p is not equal to 0.8
95 percent confidence interval:
 0.5989396 0.7854574
sample estimates:
      p 
0.7
```

The continuity correction adjusts the test statistic to account for the discrete nature of the binomial distribution, making the normal approximation more conservative.

This makes the test slightly more conservative, resulting in:

Larger p-values.

Wider confidence intervals.

Use when sample sizes are small or p is close to 0 or 1 to ensure conservative results.

For larger samples, the difference between corrected and uncorrected tests becomes negligible.

Conditions for Inference

In order for inference based on the test statistic and confidence interval formulas shown above, the following conditions need to be met:

- ▷ The **sample needs to be randomly drawn** from the population
- ▷ **The population should be at least 10 times the size of the sample.** This condition helps to ensure that the standard error of $\hat{p}$ is approximately equal to $\frac{\sqrt{p(1-p)}}{n}$.
- ▷ **The sample size n is large enough to ensure approximate normality.** For significance testing, as a general rule, we use the condition that inference is valid if $n p_0$ and $n \cdot (1 - p_0)$ are both greater than 10.
- ▷ For confidence intervals, we use the general rule that the counts of **both successes and failures are both greater than or equal to 15**. That is, $n \cdot \hat{p}$ and $n \cdot (1 - \hat{p})$ are both at least 15.

These conditions should always be checked. If the sample size is not large enough, then there are some approximations that can be used to estimate the confidence interval and adjust the z-statistic appropriately.

Two-Sample Tests for Proportions

Two-Sample Tests for Proportions

Consider situations where we are interested in comparing the **sample proportions from two populations**.

We have separate samples from each population and we want to make conclusions about the characteristics of the two populations.

We are interested in estimating the **quantity $p_1 - p_2$ or testing the null hypothesis of no difference between proportions $p_1 - p_2 = 0$ or, equivalently, $p_1 = p_2$** .

Since the population proportions are unknown, they are estimated by the sample proportions.

Population	Population proportion	Sample size	Sample proportion
1	p_1	n_1	$\hat{p}_1$
2	p_2	n_2	$\hat{p}_2$

Confidence Intervals for Differences in Proportions

The standard deviation of the difference in the sample proportions, $\hat{p}_1 - \hat{p}_2$, is given by

$$\sigma_{\hat{p}_1 - \hat{p}_2} = \sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}$$

This quantity must be estimated as it is based on the unknown parameters p_1 and p_2 . For inference which is referred to as the standard error of the statistic $\hat{p}_1 - \hat{p}_2$ we use

$$SE_{\hat{p}_1 - \hat{p}_2} = \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$$

The level C confidence interval for the difference in population proportions is given by

$$(\hat{p}_1 - \hat{p}_2) \pm z \cdot \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$$

where z is the appropriate critical value from the standard normal distribution with area C between $-z$ and z .

This confidence interval should only be used when the populations are at least 10 times as large as the samples and the counts of successes and failures are 10 or more in both samples.

An Example: Two-Sample Tests for Proportions

An investigator is interested in the long-term effects of preschool programs on low-income children. A study was conducted where by two groups of children were followed over time. The first group of 61 children did not attend preschool. The second group of 62 children (from similar areas and with similar backgrounds of those in the first sample) attended preschool as 3- and 4-year-olds. The need for social programs as adults was the outcome of interest. Of the group who did not attend preschool, 49 of them needed social services (mainly welfare) between the ages of 18 and 30. In the preschool group, 38 required social services in the same age range.

Calculate the 95% confidence interval for the difference in proportions of adults requiring social services between those who did not attend preschool versus those who did attend preschool.

*Lets define a “**success**” in this case as requiring social services.*

An Example: Two-Sample Tests for Proportions

Population	Population description	Sample size	Count of successes	Count of failures	Sample proportion
1	No Preschool	$n_1 = 61$	49	$61 - 49 = 12$	$\hat{p}_1 = \frac{49}{61} = 0.803$
2	Preschool	$n_2 = 62$	38	$62 - 38 = 24$	$\hat{p}_2 = \frac{38}{62} = 0.613$

In the table, the smallest of these quantities is 12. Since > 10 , we can proceed with using the formulas specified above.

The level C confidence interval for the difference in population proportions is given by

$$\begin{aligned} & (\hat{p}_1 - \hat{p}_2) \pm z \cdot \sqrt{\frac{\hat{p}_1(1 - \hat{p}_1)}{n_1} + \frac{\hat{p}_2(1 - \hat{p}_2)}{n_2}} \\ &= 0.190 \pm 1.960 \cdot \sqrt{\frac{0.803(1 - 0.803)}{61} + \frac{0.613(1 - 0.613)}{62}} \\ &\approx (0.033, 0.347) \end{aligned}$$

Two Sample Tests for Proportions: R commands

```
# Use the prop.test() function
> prop.test([s], [n], alternative=[alternative], conf.level=[confidence
  level],
correct=FALSE)

# [s]: number of successes in each group: c(s1, s2)
# [n]: sample size in each group: c(, )
# [alternative]: "two.sided" (default), "greater", "less"
# [confidence level]: default 0.95
# Correct = FALSE specifies not to use a continuity correction
```

Continuity correction corrects for the fact that the sampling distribution of a proportion is not a continuous distribution.

It is a very slight alteration of the formula for the test statistics and the confidence interval that makes the test a little more conservative (and thus the confidence interval a little wider). **It is especially helpful when the sample size is small.**

Two Sample Tests for Proportions: An Example

```
> prop.test(c(49, 38), c(61, 62), conf.level=0.95, correct=FALSE)
2-sample test for equality of proportions without continuity correction

data:  c(49, 38) out of c(61, 62)
X-squared = 5.383, df = 1, p-value = 0.02033
alternative hypothesis: two.sided
95 percent confidence interval:
 0.03336798 0.34738294
sample estimates:
   prop 1    prop 2 
0.8032787 0.6129032
```

```
> prop.test(c(49, 38), c(61, 62), conf.level=0.95, correct=TRUE)
2-sample test for equality of proportions with continuity correction

data:  c(49, 38) out of c(61, 62)
X-squared = 4.5027, df = 1, p-value = 0.03384
alternative hypothesis: two.sided
95 percent confidence interval:
 0.01710674 0.36364418
sample estimates:
   prop 1    prop 2 
0.8032787 0.6129032
```

Significance Tests for Differences in Proportions

We are interested testing the null hypothesis $H_0 : p_1 = p_2$.

The two-sample z-statistic used for hypothesis testing is calculated by

$$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1 - \hat{p}) \cdot \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

This quantity represents how far the difference in sample proportions is from 0 in standard deviations units.

Inference using this test is valid when the populations are at least 10 times as large as the samples and the counts of successes and failures are 5 or more in both samples.

An Example: Significance Tests for Differences in Proportions

1. Set up the hypotheses and select the alpha level

$H_0 : p_1 = p_2$ (the underlying population proportion needing adult social services among those who attended preschool is the same as that of those who did not attend preschool)

$H_1 : p_1 \neq p_2$ (the underlying population proportion needing adult social services among those who attended preschool is different than that of those who did not attend preschool)

$$\alpha = 0.05$$

2. Select the appropriate test statistic

$$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1 - \hat{p}) \cdot \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

3. State the decision rule

Determine the appropriate critical value from the standard normal distribution associated with a right hand tail probability of $\alpha/2 = 0.05/2 = 0.025$.

Decision Rule: Reject H_0 if $|z| \geq 1.960$, Otherwise, do not reject H_0

An Example: Significance Tests for Differences in Proportions

4. Compute the test statistic and the associated p-value

$$\begin{aligned} z &= \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1 - \hat{p}) \cdot \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} \\ &= \frac{0.190}{\sqrt{0.707 \cdot (1 - 0.707) \cdot \left(\frac{1}{61} + \frac{1}{62} \right)}} \\ &= 2.320 \end{aligned}$$

5. Conclusion

Reject H_0 since $|2.320| \geq 1.960$.

We have significant evidence at the $\alpha = 0.05$ level that $p_1 \neq p_2$ ($p = 0.0204$). The percentage of adults needing social services was 20% lower among those who attended preschool versus those who did not.

We reject the null hypothesis that the underlying population proportions are the same between those attending preschool versus those who did not.

Effect Measures

We focused on the differences in proportions (also known as the risk difference)

Measures of the differences between risks are called effect measures.

Effect measures include the **risk difference, the relative risk, and the odds ratio.**

The risk difference is simply the absolute difference in risk between two populations.

$$\text{Estimate of risk difference} = \widehat{RD} = \hat{p}_1 - \hat{p}_2$$

- ▷ A risk difference of 0 indicates that there is no difference in risk between the populations.
- ▷ A positive risk difference indicates a higher risk in group 1
- ▷ A negative risk difference indicates a lower risk in group 1
- ▷ Generally, group 2 as defined here is thought of as the reference group

Estimate of risk ratio

$$\text{Estimate of risk difference} = \widehat{RD} = \hat{p}_1 - \hat{p}_2$$

The relative risk is simply the ratio of the risks in each of the two populations.

$$\text{Estimate of risk ratio} = \widehat{RR} = \frac{\hat{p}_1}{\hat{p}_2}$$

- ▷ A relative risk of 1 indicates that **there is no difference in risk between the populations.**
- ▷ A relative risk **greater than 1 indicates a higher risk** in group 1
- ▷ A relative risk **less than 1 indicates a lower risk** in group 1

Effect Measures - Odds Ratio

The odds ratio is the ratio of the odds of the outcome in each of the two populations. The odds of an event is the ratio of the events to the non-events (or it is the ratio of the number of successes to the number of failures).

If among n subjects in a particular group, x had the event, then odds of the event in that group would be $\frac{x}{n-x}$ which is equal to $\frac{\hat{p}_1}{1-\hat{p}_1}$.

The odds ratio, then is estimated as:

$$\text{Estimate of odds ratio} = \widehat{OR} = \frac{\frac{\hat{p}_1}{1-\hat{p}_1}}{\frac{\hat{p}_2}{1-\hat{p}_2}}$$

Estimate of odds ratio

$$\text{Estimate of odds ratio} = \widehat{OR} = \frac{\frac{\hat{p}_1}{1-\hat{p}_1}}{\frac{\hat{p}_2}{1-\hat{p}_2}}$$

- ▷ An odds ratio of **1** indicates that there is no difference in odds between the populations
- ▷ An odds ratio **greater than 1** indicates a higher odds in group 1
- ▷ An odds ratio **less than 1** indicates a lower odds in group 1
- ▷ **When the event is rare (when are small), the risk ratio and the odds ratios are similar in magnitude.**

Odds ratios **are less intuitive to interpret**, but have some favorable mathematical properties and as such are used quite often in statistics.

An Example: Estimate of risk difference

Suppose we are interested in the association between gender and risk of having a coronary event in a high-risk patient population (who have had an event in the past). We have data for 25 females and 25 males. We followed each subject for a year to see if they had another coronary event (defined here as a “success”).

Calculate the estimated risk difference.

Population	Population description	Sample size	Count of “Successes”	Count of “Failures”	Sample proportion
1	Males	$n_1 = 25$	18	$25 - 18 = 7$	$\hat{p}_1 = \frac{18}{25} = 0.72$
2	Females	$n_2 = 25$	8	$25 - 8 = 17$	$\hat{p}_2 = \frac{8}{25} = 0.32$

Estimate of risk difference = $\widehat{RD} = \hat{p}_1 - \hat{p}_2 = 0.72 - 0.32 = 0.40$

The risk of a coronary event is 40% higher among males as among females.

An Example: Estimate of risk ratio

$$\text{Estimate of risk ratio} = \widehat{RR} = \frac{\hat{p}_1}{\hat{p}_2} = \frac{0.72}{0.32} = 2.25$$

The risk of a coronary event is 2.25 times higher among males as among females.

If the reference group had been males, the estimate of the relative risk would have been

$$\text{Estimate of risk ratio} = \widehat{RR} = \frac{\hat{p}_1}{\hat{p}_2} = \frac{0.32}{0.72} = 0.44$$

The risk of a coronary event among females is less than one half the risk among males.

Sometimes the actual result and the wording of the interpretation help dictate the choice of the reference group.

Here, “the risk of a coronary event is 2.25 times higher among males as among females” is easier to understand than “the risk of a coronary event among females is less than one half ($\hat{RR} = 0.44$) the risk among males.”

An example: Estimate of odds ratio

$$\text{Estimate of odds ratio} = \widehat{OR} = \frac{\frac{\hat{p}_1}{1-\hat{p}_1}}{\frac{\hat{p}_2}{1-\hat{p}_2}} = \frac{2.57}{0.47} = 5.46$$

The odds of a coronary event is 5.46 times higher among males as among females.

If the reference group had been males,

$$\text{Estimate of odds ratio} = \widehat{OR} = \frac{\frac{\hat{p}_1}{1-\hat{p}_1}}{\frac{\hat{p}_2}{1-\hat{p}_2}} = \frac{\frac{0.32}{1-0.32}}{\frac{0.72}{1-0.72}} = \frac{0.47}{2.57} = 0.18$$

The risk of a coronary event among females is less than one half the risk among males.

As noted above, sometimes the actual result and the wording of the interpretation help dictate the choice of the reference group. “*The odds of a coronary event is 5.46 times higher among males as among females*” is easier to understand than “*The odds of a coronary event among females is less than one fifth the odds of the same event among males.*”